A Finite Relational Complex-Structure Seed and Its Nonselection Boundaries

Conditional Algebra, Projective Phase and Operational Countermodels

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Research direction and responsible authorship

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Initial External Grounding: open; no quantum/classical physical emergence established

Abstract

We test whether an exactly specified *finite, timeless directed relation* can supply a complex structural ingredient without inserting complex amplitudes into its primitive data. A three-cycle, functorially represented on a rational zero-sum module, gives an integer skew endomorphism C obeying $C^2 = -3I$ on that module; hence its rational endomorphism algebra contains an imaginary quadratic field. Directed reversal changes C to $-C$, while retaining the full unoriented permutation symmetry rules out an equivariant complex structure. After *additional, explicitly imported* real completion, duplication, norm and circle quotient, the same construction provides the standard geometry of $\mathbb{CP}^1$ and a nonreal Bargmann triple. But the original three-cycle also supports an ordinary classical probability simplex. Therefore the finite algebraic seed selects neither the Born rule nor quantum composition, dynamics or classical mechanics. An exact three-overlap Gram feasibility theorem and a tightly delimited topological symplectic discriminator sharpen the negative and conditional positive findings. This is mathematical structural research, *not* an ontology or a physical correspondence.

1 A rational structure generated by a finite relation

The *only initial combinatorial datum* for the following trial is the directed relation $1 \to 2 \to 3 \to 1$, represented by its permutation P on the free integer module $\mathbb{Z}^3$. This representation choice is an explicit mathematical *linearization functor*, not a law forced by an observation. Neither P nor its arrows is interpreted as a physical update or time step. Write

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad C = P - P^{\mathsf{T}}, \quad Q = I - \tfrac{1}{3}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \quad V_{\mathbb{Q}} = \{x \in \mathbb{Q}^3 : \mathbf{1}^{\mathsf{T}} x = 0\}.$$

Theorem 1 (Exact complex-type algebraic seed)**.** *On $V_{\mathbb{Q}}$, C is invertible and satisfies*

$$C^2 = -3I_{V_{\mathbb{Q}}}, \qquad \mathbb{Q}[C|_{V_{\mathbb{Q}}}] \simeq \mathbb{Q}[t]/(t^2+3) \simeq \mathbb{Q}(\sqrt{-3}). \tag{1}$$

The alternating form $\Omega(x,y) = x^{\mathsf{T}} C y$ is nondegenerate on $V_{\mathbb{Q}}$. After choosing *scalar extension to $\mathbb{R}$ and the standard counting inner product, $J = C/\sqrt{3}$ is an orthogonal complex structure on $V_{\mathbb{R}}$ and Ω defines a compatible signed area form up to positive scale.*

Proof. The permutation has $P^3 = I$ and $P^{\mathsf{T}} = P^2$. Since $P + P^2 = \mathbf{1}\mathbf{1}^{\mathsf{T}} - I$, we have $(P - P^2)^2 = P + P^2 - 2I = -3Q$. The projector Q acts as identity on $V_{\mathbb{Q}}$ and $C^{\mathsf{T}} = -C$. The polynomial $t^2 + 3$ is irreducible over $\mathbb{Q}$, proving the field assertion. The vector-pair $v_1 = (1, -1, 0)^{\mathsf{T}}$, $v_2 = (0, 1, -1)^{\mathsf{T}}$ spans the integer zero-sum lattice; direct evaluation gives $\Omega(v_1, v_2) = -3$, so Ω is nondegenerate. Finally $J^2 = -I$ and $J^{\mathsf{T}} = -J$ on the real zero-sum plane. This proves the algebraic claims without any complex scalar at the input. □

Theorem 2 (Reflection obstruction and permitted symmetries)**.** *Let R exchange vertices 1 and 2. Then $RPR^{-1} = P^{-1}$ and $RCR^{-1} = -C$. The commutant on $V_{\mathbb{R}}$ of the* oriented *cyclic action is $\mathbb{R}I + \mathbb{R}J \simeq \mathbb{C}$. The commutant of the* unoriented *full permutation action is only $\mathbb{R}I$; consequently it contains no real linear J' with $(J')^2 = -I$.*

Proof. The reversal relation follows by multiplication. On the zero-sum plane, P has the irreducible real minimal polynomial $t^2 + t + 1$, hence its two-dimensional centralizer is $\mathbb{R}[P] = \mathbb{R}I + \mathbb{R}J$. Requiring a commuting map $aI + bJ$ also to commute with R forces $b = 0$, because $RJR^{-1} = -J$. A real scalar squared cannot equal $-I$. □

Relabeling caveat. Reversing a finite directed cycle gives an *isomorphic* unlabelled directed graph after reversing vertex labels. The construction is covariant with the labelled oriented relation; it does not produce a physically measurable absolute chirality or distinguish conjugate descriptions without an independent reference. If the input is only an unoriented triangle, J is not symmetry-canonical.

2 A complete mathematical positive control, with its imports exposed

Let $W = V_{\mathbb{R}} \oplus V_{\mathbb{R}}$. Choose the inherited counting inner products on both summands, choose the diagonal action of J, and *then* define W as a complex two-dimensional Hermitian vector space with $i \cdot w = Jw$. Choose a unit sphere $S^3 \subset W$, the continuous circle action e^{tJ} and its ray quotient:

$$\left(S^3/\{e^{tJ} : t \in \mathbb{R}\}\right) \;\cong\; \mathbb{CP}^1. \tag{2}$$

This is the usual Hopf-projective/symplectic-reduction construction; the Fubini–Study metric and area form arise *from the chosen Hermitian completion.* No topology or *physical* probability was derived directly from the three vertices. The choice of two copies, continuous scalar completion, circle quotient and measurement rule are substantial added premises.

For an explicit *conditional* phase witness, choose orthonormal complex vectors e, f in W, and let $\zeta = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, the action of P on a selected complex representation. Put

$$x = e, \qquad y = (e + f)/\sqrt{2}, \qquad z = (e + \zeta f)/\sqrt{2}.$$

The usual inner product and the *stipulated* rule $p_{uv} = |\langle u, v\rangle|^2$ give

$$(p_{xy}, p_{yz}, p_{zx}) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{2}), \qquad B_{xyz} = \langle x, y\rangle\langle y, z\rangle\langle z, x\rangle = \frac{1 + i\sqrt{3}}{8}. \tag{3}$$

Reversing the cycle conjugates this Bargmann value. Thus the finite directed relation can *seed* a standard nontrivial complex representation, but the physical interpretation of its complex phase remains unwarranted.

Proposition 3 (Same finite seed, conventional classical completion)**.** *The same original three labels and cyclic permutation support the classical operational theory $\Delta_2 = \{p \in \mathbb{R}^3 : p_j \geq 0,\ \sum p_j = 1\}$, with ordinary affine effects and the reversible label permutation P. Its zero-sum contrast representation is exactly $V_{\mathbb{R}}$ with the same $C^2 = -3I$. Hence the existence of this algebraic complex structure alone does not select quantum mechanics.*

Proof. The simplex and its effects give the classical linear expectation $E_p(f) = \sum_j f_j p_j$. The linear action of P on differences $p - q$ preserves $V_{\mathbb{R}}$ and therefore inherits the foregoing identity. Its pure states are the three vertices, not the continuum of quantum rays, and its effects do not supply nonreal Bargmann measurement phases. This is a directly checked pair of inequivalent completions of the same initial combinatorial datum. □

3 Exact feasibility of a three-ray phase

Theorem 4 (Three-overlap feasibility, and the dimension boundary)**.** *Fix $0 < p_{12}, p_{23}, p_{31} \leq 1$, let $s = p_{12}+p_{23}+p_{31}$, $r = \sqrt{p_{12}p_{23}p_{31}} > 0$, and set $\kappa = (s - 1)/(2r)$. For three normalized complex rays with these squared pairwise overlaps and Bargmann phase ϕ:*

- *a realization in some $\mathbb{C}^d$ with $d \geq 3$ exists* if and only if $\cos\phi \geq \kappa$*;*
- *a realization in $\mathbb{C}^2$ exists* if and only if $\cos\phi = \kappa$*.*

When the right-hand condition is possible, three pairwise squared overlaps determine the qubit Bargmann phase up to conjugation*, but can leave a continuous range for a qutrit.*

Proof. Rephase representatives so that their Gram matrix has entries

$$G = \begin{pmatrix} 1 & a & \bar{c} \\ a & 1 & b \\ c & b & 1 \end{pmatrix}, \quad a = \sqrt{p_{12}},\ b = \sqrt{p_{23}},\ c = \sqrt{p_{31}}e^{i\phi}.$$

Then $\det G = 1 - s + 2r\cos\phi = 2r(\cos\phi - \kappa)$. The one-by-one and two-by-two principal minors are nonnegative by hypothesis, so G is positive semidefinite precisely when $\det G \geq 0$; every positive semidefinite 3×3 Gram matrix admits a realization in $\mathbb{C}^3$. Realization in $\mathbb{C}^2$ additionally requires $\operatorname{rank} G \leq 2$, equivalently $\det G = 0$ here. The triangle product of the specified overlaps is $re^{i\phi}$. □

Exact diagnostic. If $p_{12} = p_{23} = p_{31} = 1/4$, then $\kappa = -1$ and

$$\det G = \frac{1 + \cos\phi}{4}.$$

Every phase ϕ is possible for a complex qutrit; a qubit can realize only $\phi = \pi \pmod{2\pi}$ (the rank-two endpoint). The phase range for a *chosen triple* must not be conflated with ambiguity of the entire full projective manifold, for which transition-probability-preserving bijections have much stronger structure. This exact Gram relation and the conjugation obstruction are consistent with the published qubit result of Li, Wagner and Zhang [4].

4 A limited topological discriminator, not a physics selection

Theorem 5 (Conditional single-system symplectic classification)**.** *Within the* finite-dimensional, simple Euclidean Jordan state-space list *(equivalently, the strongly symmetric spectral class modulo classical simplices), suppose the pure-state manifold is compact, connected, smooth and of positive dimension. Then it admits a symplectic form precisely in the complex-projective cases, counting the three-dimensional Bloch ball as the standard complex two-level quantum state space.*

Proof. The listed nonclassical irreducible pure manifolds are real, complex and quaternionic projective spaces, the octonionic projective plane, and spheres from spin factors [2, 3]. For a compact positive-dimensional symplectic manifold, $[\omega] \neq 0 \in H^2_{\mathrm{dR}}$ by Stokes' theorem, since ω^n is a volume form. Positive-dimensional $\mathbb{RP}^k$ and $\mathbb{HP}^k$ have zero real H^2, as does $\mathbb{OP}^2$; for spheres S^m with $m \geq 1$, $H^2(S^m;\mathbb{R}) \neq 0$ only for $m = 2$. Conversely, $\mathbb{CP}^k$ carries its Fubini–Study symplectic form, and $S^2 \cong \mathbb{CP}^1$ is the 3-ball spin factor's pure manifold. Classical simplex pure-state sets are discrete and were excluded by the positive-dimensional hypothesis. These are standard topological facts [5]. □

Premise boundary. This is a valid *conditional discriminator inside a previously imposed finite-dimensional Jordan/strongly-symmetric spectral class*, not a theorem for arbitrary operational theories. Barnum, Müller and Ududec's original 2014 characterization explicitly required classical decomposability, strong symmetry and no higher-order interference [2]. Barnum and Hilgert's later 2020 theorem obtains the Jordan list from the first two conditions alone, removing the higher-order-interference assumption for that classification [3]. Neither postulate package has been derived from the prephysical relation. Classical spin S^2 with arbitrary Borel probability measures supplies an unrestricted counterexample to interpreting symplectic pure geometry as uniquely quantum. This contrast cannot be suppressed by the narrower classification.

5 Finite-carrier and operational nonselection checks

Gelander's published theorem rules out $\mathbb{CP}^n$ as a Gromov–Hausdorff limit of finite *exactly homogeneous metric spaces* [1]. The derived representation in Section 2 does not contradict that result: it neither exhibits a sequence of finite homogeneous metric carriers approaching $\mathbb{CP}^1$ in the Gromov–Hausdorff sense nor derives a metric from a prescribed physical experiment. It uses a finite relation to construct a real vector representation and then applies

additional continuous operations. This alternative route is a *mathematical trial*, not a demonstrated escape to physical emergence.

Even if the *entire* single-system complex projective geometry and its holonomy were somehow identified, a composition rule remains necessary. As shown by the separately reproduced minimal generalized-probabilistic-theory tensor-product control, separable-only and full quantum composite state spaces agree on local qubit data but disagree on entangled preparations and CHSH correlations [6]. Choosing the standard product Hilbert space is a new premise. The same static local geometry also permits inequivalent Hamiltonian choices and no calibrated clock. Separately, loss of phase coherence under a stochastic channel does not prove convergence to reversible classical Hamiltonian flow.

6 Typed status and prospective tests

Statement	Status	Missing or checked restriction
$C^2 = -3I$ on rational contrast space; mirror obstruction	Exact, supported in declared model	The free-module functor is a selected mathematical representation
$S^3/U(1) \cong \mathbb{CP}^1$ from doubled real plane	Conditional construction	Real completion, norm, duplication, circle action and quotient imported
Bargmann triple and Gram feasibility theorem	Exact conditional algebra	Complex rays and squared-overlap rule stipulated
Symplectic selector for the simple Jordan list	Conditional mathematical selection	Jordan/spectral/strong-symmetry premises and physical operational access absent
Full complex quantum statistics and entangling composites	Not selected	Classical simplex and minimal-composite witnesses
Physical time, calibrated instruments, space, classical Hamiltonian limit	Open	No primitive-to-experiment correspondence or controlled dynamics map
Total ontology	Not identified	Quantum and classical emergence, full empirical map incomplete

The immediate falsifiable mathematical search contract is to specify a *non-stipulated* map from independently defined finite relations to operational probabilities and transformations, then compare its predictions against the classical-simplex completion and against conjugate complex completions. A nontrivial new test must discriminate those completions with a declared measurement interface, controls, errors and alternative mechanisms. Successful internal numerical geometry tests alone do not count. No physical hypothesis, empirical measurement or independent peer review is claimed by this note.

References

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